Multihop LoRa OTA Firmware Updates for APRS Networks

Experimental results, architecture, and further development
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1. Objective

Update ESP32 LoRa APRS iGates and digipeaters over RF, without an IP connection or physical intervention at the remote target. The longer-term goal is to distribute an image through several relays to one or more devices, with recovery after interruption and restoration of normal APRS operation.

An initial implementation of controlled distribution has now passed laboratory testing: one controller, one relay, and one target, using block groups, local repair, and recovery after interruption. Subsequent short tests checked relay restarts, a real power cut with a partial page, and an RF-selected temporary profile. Multiple paths and distribution to several targets remain to be developed.

2. Test results

The tests used TTGO LoRa32 V2.1 boards with ESP32 processors and SX1278 radios. The relayed setup is **F4MLV-2 → F4MLV-10 → F4MLV-15**. The prototype is integrated into the iGate firmware containing RXT, native-frame support, and standalone PPM decoding.

The fast profile used for the complete-image tests on 30 October is **BW500 kHz, SF7, CR4/5, 2 dBm**, giving **21.875 kbps nominal**. The initial bench profile is 434.100 MHz, BW125/SF12/CR4/5; the fast profile uses 434.500 MHz. These settings are specific to the laboratory setup.

Test	Image, bytes	Transfer and verification	Useful throughput
Direct link, target Wi-Fi off	1,744,480	26 min 38 s	1,092 bytes/s
One relay, ACK per block	1,760,528	1 h 08 min 18 s	430 bytes/s
One relay, groups and persistent resume	1,764,720	53 min 24 s	551 bytes/s

Durations cover firmware data transfer and image verification, not initial setup, initial profile negotiation, activation, or boot confirmation. The final test includes **73.24 seconds of deliberate recovery after a restart during the transfer**.

In the relayed setup, grouped transfer reduced the observed duration by about **22%** and increased useful throughput by about **28%**. The images are similar but not identical in size. This compares two successive tests, not a guaranteed improvement for every network or a direct measurement of cumulative airtime.

The target had both Wi-Fi and its access point disabled. Authenticated RF replies checked this state before and after the update. Relay Wi-Fi remained enabled for management. Firmware blocks and target commands travelled over RF. Initial provisioning and independent flash readback used USB.

3. What was verified

The complete image was written to the inactive ESP32 partition, checked using SHA-256, and validated as an ESP32 application image. An independent USB readback of that partition matches the transmitted file byte for byte.

Activation remains a separate authenticated command. Two returns to the previous firmware were observed: after an injected reset before confirmation, and after activation with confirmation withheld beyond the deadline. The new firmware was then activated and confirmed over RF. After another restart, it still answered through the relay with Wi-Fi off.

Switching to the fast profile and returning to the initial profile worked. Checks before the transfer also showed no target response to a direct controller command or while the relay was paused; replies resumed when the relay was enabled.

APRS text, native, and PPM reception had been checked in earlier direct-link tests. Those reception checks were not repeated in the final grouped-transfer test.

4. How grouped transfer works

The controller sends a group of **32 blocks of 128 bytes**, forming a **4,096-byte flash page**, without waiting for an ACK after each block. The final page may be shorter.

The relay listens to the group and caches the signed frames in RAM. On the controller's request, it forwards them to the target and asks for a reception summary containing a 32-bit bitmap. The controller and relay therefore take turns transmitting.

If the target is missing blocks held in the relay's cache, the relay repairs them locally. Remaining holes are reported to the controller, which sends only the blocks still missing. The cache holds **one group**, not an entire image, and is lost when the relay restarts.

Over the complete transfer, the relay counted **94 local block repairs** and the controller **17 requested block retransmissions**. One loss had been deliberately introduced on each link. These counters are neither an RF loss rate nor a diagnosis of the cause of losses.

5. Persistent resume: what survives

The target accepts out-of-order blocks within the current group and checks duplicates. It writes a complete page to the inactive partition. Only then does an NVS checkpoint record the image identity and durable offset. Blocks from an incomplete page remain in RAM.

During the complete transfer on 30 October, the target committed **8,192 bytes**, then received **2,048 additional bytes** without completing the next page. An RF diagnostic command reset the target. After the nodes returned to their initial profiles and a fresh authenticated session was established, the target recovered the **8,192 committed bytes**. Only the incomplete page was retransmitted; committed pages were not sent again.

Session expiry preserves this progress; explicit cancellation clears it. Resumption uses fresh authentication rather than reusing the old nonce.

Targeted tests on 31 October

Each test below continued to **16,384 bytes**, independently read back from flash and identical to the file. These short tests complement the previous complete transfer; they are not additional complete-image transfers.

Interruption	Recovered progress	Observed result
Relay restarted while its cache was being loaded	8,192 bytes	Interrupted page retransmitted
Relay restarted after sending 16 blocks towards the target	8,192 bytes	Interrupted group retransmitted
Relay restarted before forwarding the ACK of a committed page	12,288 bytes	Committed page not retransmitted
Target USB power removed for 22.22 seconds with a partial page	8,192 bytes	Partial page rewritten in full

For the physical power cut, a diagnostic had erased a 4,096-byte page and written its first half without advancing the checkpoint. A readback after power-up, before resuming, found the expected **2,048 programmed bytes and 2,048 erased bytes**. Resumption preserved committed pages and rewrote the incomplete page.

Activation of incomplete images was rejected and the running application remained selected. **Power was removed between programming operations. A cut during an actual erase, program operation, or NVS checkpoint write remains to be tested.**

6. The two project architectures

Architecture 1 — Controlled distribution

A controller selects participants and schedules block delivery. Relays can cache blocks; receivers report holes for selective repair.

The prototype now implements an initial version: **one fixed path, one relay, one target, and a one-group cache**. Shared distribution to several targets, larger caches, and chains of several intermediate relays have not yet been implemented.

Architecture 2 — Multiple paths to a target

Two or more paths could maintain delivery when a link becomes unavailable. This requires coordinated transmissions, duplicate handling, and separate path state.

This extension remains to be developed. It supplements controlled distribution rather than replacing it. The recommendation remains to use Architecture 1 as the foundation and add alternate paths where their benefit is demonstrated.

7. Control and profile switching

Preparation, profile changes, activation, and confirmation use authenticated binary OTA commands. “Normal APRS profile” means the stored radio settings, not commands carried as ordinary APRS messages. The bench temporarily uses its own initial profile.

The controller prepares the session on that initial profile. The target acknowledges switching; the relay forwards the acknowledgment, then each node switches to the fast profile after its local delay. The controller checks the link before sending data. After image verification, participants return to the initial profile for activation and confirmation.

A node with one radio cannot keep listening to the initial profile while using the fast profile. Return commands during transfer therefore use the fast profile. A local inactivity lease restores stored settings if the link disappears; recovery does not depend on receiving a final command. A restart

also reloads stored radio settings.

Controller-selected temporary profile

The authenticated RF command can now carry **frequency, bandwidth, SF, CR, and requested transceiver power**. Target and relay check compatibility with their radios before switching. Settings remain in RAM; normal saved settings are not overwritten.

The test used **434.550 MHz, BW250, SF8, CR4/6, requested power 3 dBm**, instead of the first prototype's fixed bench profile. Four pages were transferred and read back identically. An SF5 request was rejected by the SX1278 relay. Return to the normal profile was checked after silence, by command, and after reset. RF output power was not measured.

For now, the controller reads a `normal` / `transfer` JSON file supplied to the sending tool with `--radio-profiles`, and sends the temporary settings over RF. Signed APRS commands are now implemented to stage the profile, query status, switch temporarily, and restore the normal profile. A separate APRS test checks replies, profile staging, and the logged switch and automatic return 30.001 seconds later. On 1 November, direct-link APRS exchanges on the temporary profile and explicit, inactivity, and reset returns were checked on hardware. That test also checks RXT tuples present with ON, absent with OFF, and explicit saving of the setting across restarts. They control one node, without coordinating several relays or starting the image transfer. The new firmware also accepts the old profile selector.

This mechanism works for the tested route. Coordinating several relays or different profiles per link is a separate extension.

RF commands from Graywolf

Enter these in the **authenticated command field**, without prepending `?` or `@@#`. Graywolf builds the signed `!RC1` envelope using the target key and counter. The signed identity is the board's Callsign-SSID; the RF addressee is its tactical callsign if set, otherwise its Callsign-SSID.

Command to sign	Effect
`RXT=ON`	Enables appending local measurements to relayed RF packets.
`RXT=OFF`	Disables this addition; decoding and forwarding measurements already present remain active.
`RXT=?`	Returns `RXT=ON` or `RXT=OFF`.
`OTA=?`	Returns the mode, session arming state, and staged temporary profile.
`OTA=PROFILE,434550000,250000,8,6,3`	Stages in RAM: 434.550 MHz, BW250, SF8, CR4/6, requested power 3 dBm. **Does not switch the radio yet.**
`OTA=FAST,30`	Switches to the staged profile two seconds after transmitting the reply; inactivity lease of 30 seconds.
`OTA=NORMAL`	Replies on the current profile, then restores the normal profile; suspends any session while retaining the durable checkpoint.

The existing **`COMMIT`** command, also signed, saves the RXT setting across restarts. It does not make the temporary OTA profile permanent.

For `OTA=PROFILE`, frequency and bandwidth are integer Hz; the last three values are SF, CR denominator, and requested power in dBm. `OTA=FAST` accepts leases from 15 to 60 seconds. An active session prevents staging a new profile or starting another APRS profile switch.

After `OTA=FAST`, the controller radio must use the same temporary profile to reach the target. Graywolf does not change that radio's settings itself. If it remains on the normal profile, wait for the lease to expire and the target to return automatically. Send `OTA=NORMAL` on the temporary profile for an explicit return. These commands do not start image transfer or coordinate several relays switching together.

8. Authentication, integrity, and boot

The protocol already includes controller and target identities, a persistent arming counter, a session nonce, block offsets and lengths, an authenticated manifest, and HMACs. The relay envelope adds the transmitting hop's identity and separate transport authentication; the relay preserves the target's end-to-end authenticator unchanged.

The manifest identifies image size, SHA-256, and the expected laboratory board. SHA-256 checks the content; manifest authentication protects the expected digest. The ESP32 image is checked before activation, and selecting the new firmware remains separate from receiving it.

RF boot confirmation and application rollback are independent of download resumption. An incomplete image must never be selected for boot. Extending this to other boards and partition layouts requires their own compatibility checks.

9. Hardware and data rates

BW500/SF7/CR4/5 can provide a common profile for SX1278 and SX1262 devices. The tests described here used SX1278 radios only.

SX1262 supports SF5: BW500/SF5/CR4/5 gives **62.5 kbps nominal**. SX1278 does not support SF5, so that rate has not been tested. Different profiles per link could optimize a mixed network, but they are not required to start with a compatible common profile.

These physical rates are not useful firmware throughput: headers, two hops, transmit/receive changes, reception summaries, and repairs all contribute to transfer duration.

10. Initial installation at remote sites

The OTA path inspected in Ricardo's firmware uses ElegantOTA and the ESP32 Web server. It carries firmware over IP and does not receive this protocol's blocks over LoRa.

An existing node must therefore first receive compatible firmware through Web OTA where reachable or local flashing. The experimental prototype described here has not been published as a deployment release for isolated sites.

11. Validation status

Topic	Status
One relay, complete image, target Wi-Fi off	Validated in the laboratory
Groups, cache, and selective repair	Validated over the full transfer
Resume after target reset	Validated on hardware
Real power cut with a partially written page	Validated in a short test
Cut during an actual flash/NVS operation	Simulated on host; hardware test pending
Relay restart during transfer	Three injection points checked in short tests
Configurable temporary profile over RF	Validated on four pages and return checks
Control through authenticated APRS messages	Temporary-profile exchanges and explicit, inactivity, and reset returns checked on a direct link
Several successive relays	Still to implement and test
Alternate paths and multiple targets	Still to implement and test
SX1262/SF5 and different profiles per link	Not measured
Remote sites, hidden nodes, asymmetry, busy channel	Field tests pending

Cuts during actual write operations, path handling, and coexistence with APRS traffic still need testing before treating this as ready for inaccessible sites.

12. Current position

The first working form of Architecture 1 has been demonstrated on a two-hop path. Grouped transfer, local repair, and persistent resume work together over a complete image, with verified integrity, separate activation, RF confirmation, and rollback.

The targeted tests add relay-failure recovery, a real power cut with a partial page, and controller-selected RF profiles. Next work concerns coordination of multiple devices, cuts during actual write operations, field testing, and then extension to several relays, paths, and targets. The overall project architecture has not yet been fully validated.

The experimental code, 30 October complete transfer, fault-recovery tests, and RF-selected profiles are public. Detailed test reports are in French. Code and reports for 31 October are published in commit `68707c5`; the complete transfer on 30 October used `8241529`. This publishes experimental sources and results, not a deployment firmware release for isolated sites.

The 1 November APRS and RXT tests supplement these results; they do not include an image transfer.